

Computer Networks

Transmission Media (Guided)

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□ Guided medium:

- Twisted pair, coaxial cable, optical fiber
 - **Twisted pair and coaxial cable:** Use metallic (copper) conductors that accept and transport signals in the form of electric current
 - **Optical fiber** is a cable that ; ports signals in the form of light
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Twisted pair cable

□ Twisted pair cable:

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together
- Two wires carry signals of opposite polarities
- The receiver uses the difference between the two



Twisted pair cable

□ Twisted pair cable:

- Twisting makes sure that both wires are equally affected by external influences (noise or crosstalk)
 - Converse is also true



□ Twisted pair cable:

- Unshielded Twisted pair

- ordinary telephone wire
- cheapest
- easiest to install
- suffers from external electromagnetic interference

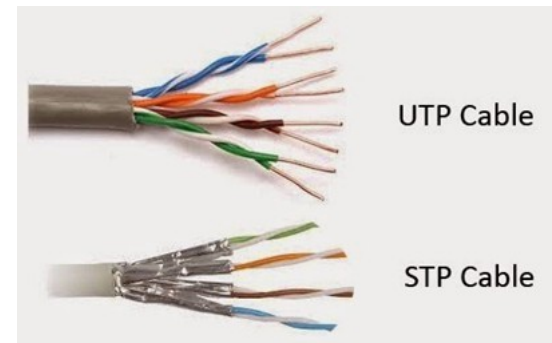
- Shielded Twisted pair:

- Has a metal foil or braided mesh covering that encases each pair of insulated conductors
- Improves the quality of cable by preventing the penetration of noise or crosstalk
- Bulkier and more expensive

- Used in telephone lines and LANs

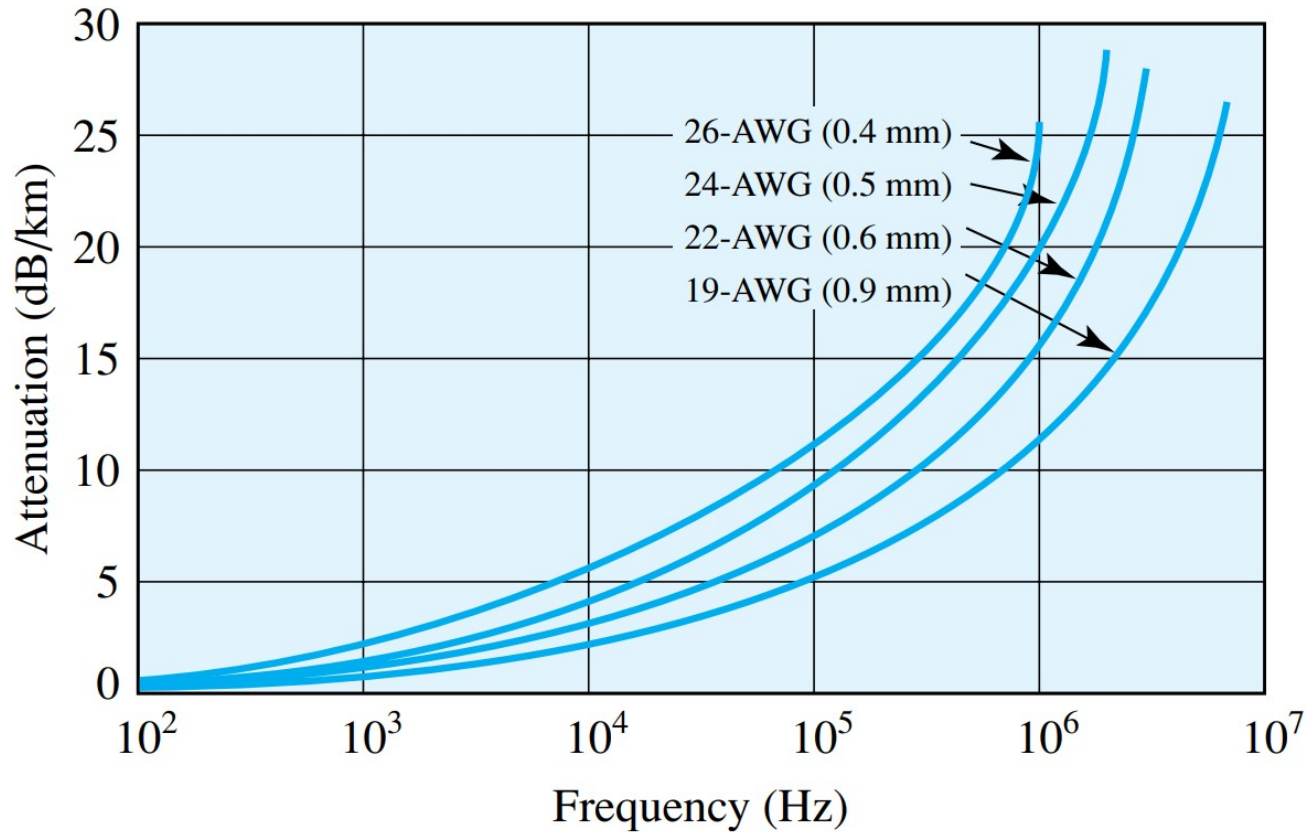


Src: https://commons.wikimedia.org/wiki/File:UTP_cable.jpg



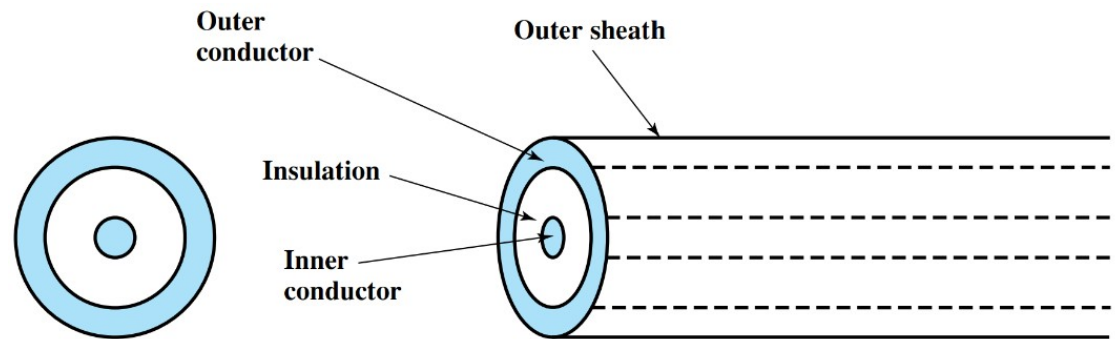
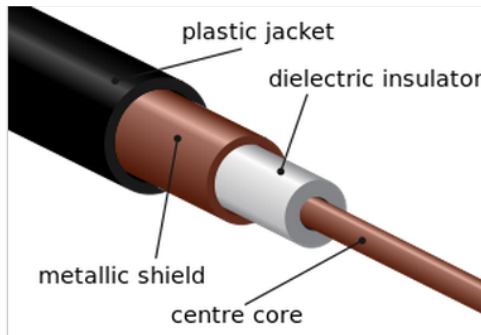
Src:
https://medium.com/@bilby_yang/comparison-between-utp-and-stp-27f7ac1d61aa

Twisted Pair Cable Performance



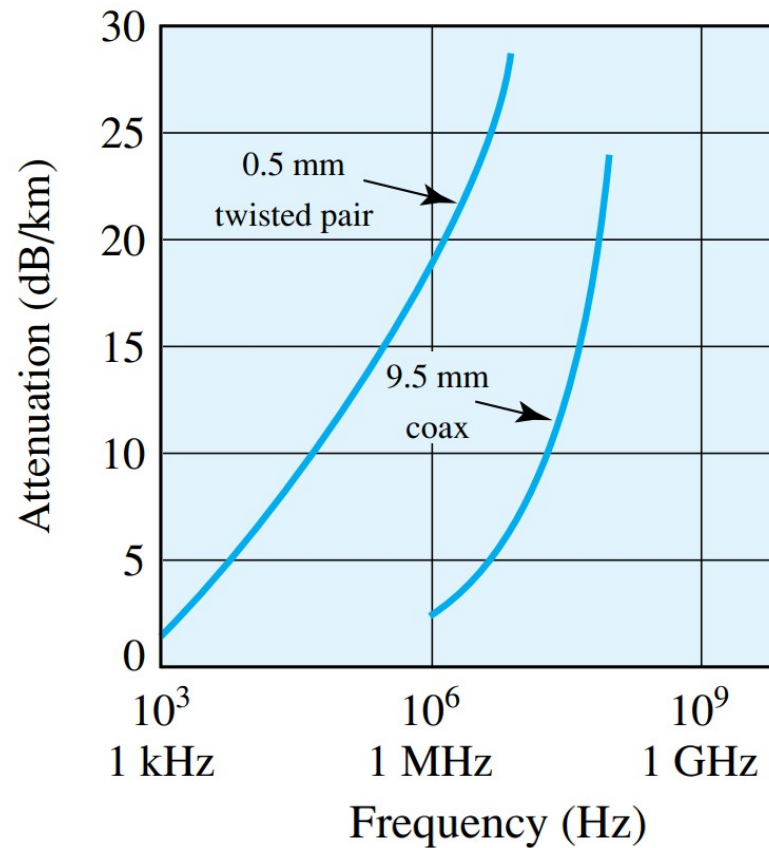
Coaxial Cable

- ❑ Coaxial cable can be used over longer distances
- ❑ Supports more stations on a shared line than twisted pair
- ❑ Consists of a hollow outer **cylindrical conductor** that surrounds a single **inner wire conductor**
- ❑ Used for TV distribution, long distance telephone transmission and LANs



Src:https://commons.wikimedia.org/wiki/File:Coaxial_cable_cutaway.svg

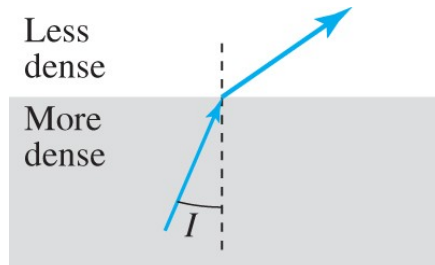
Coaxial Cable Performance



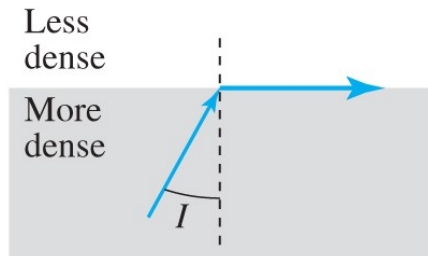
Optical Fiber

□ Uses total internal reflection to transmit light

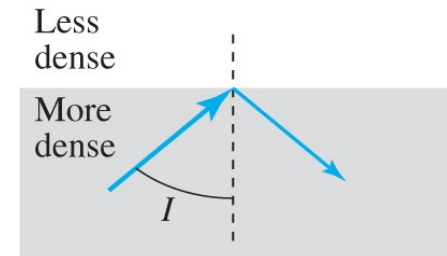
- When waves are refracted from a dense medium to a lighter medium the **angle of refraction** is greater than the **angle of incidence**
- As the angle of incidence approaches a certain threshold (called the **critical angle**), the angle of refraction approaches $90^\circ \rightarrow$ the refracted ray becomes parallel to the boundary surface
- As the angle of incidence increases beyond the critical angle, the conditions of refraction can no longer be satisfied, so there is no refracted ray, and the partial reflection becomes total



$I < \text{critical angle,}$
refraction



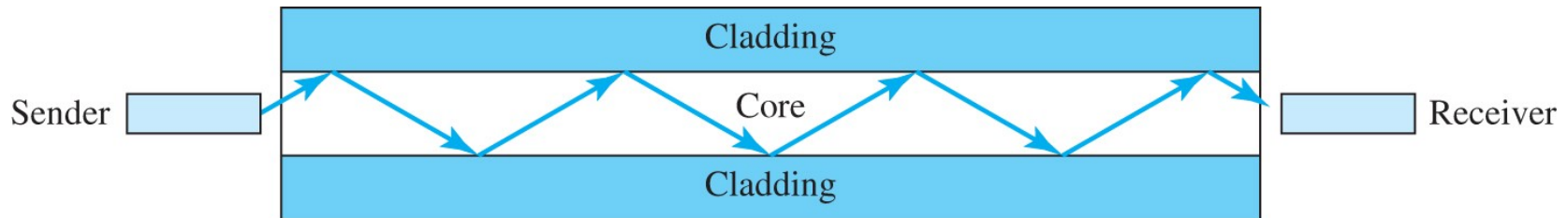
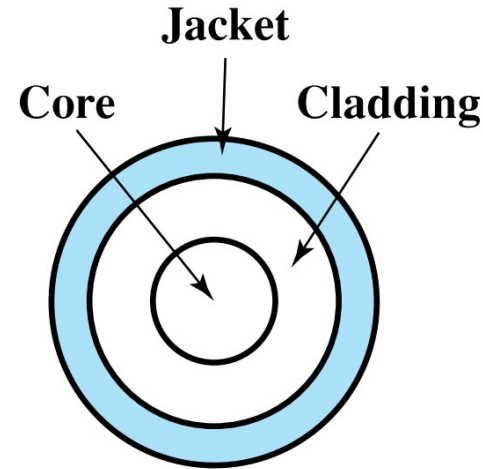
$I = \text{critical angle,}$
refraction



$I > \text{critical angle,}$
refraction

Optical Fiber

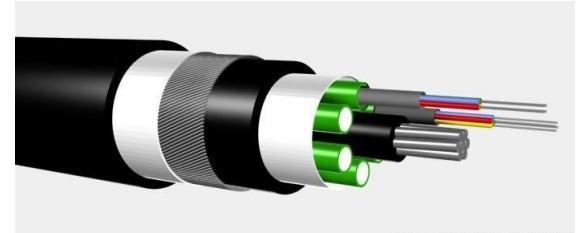
- ❑ Optical fiber is a thin flexible medium capable of guiding an optical ray
- ❑ Various glasses and plastics can be used to make optical fibers
- ❑ Has a cylindrical shape with three sections
 - Core, cladding, jacket



Optical Fiber

□ Light sources used:

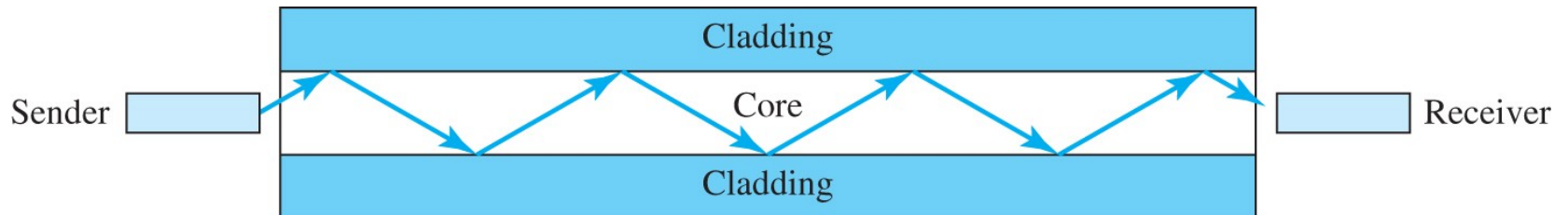
- Light Emitting Diode (LED)
- Cheaper, operates over a greater temperature range, lasts longer



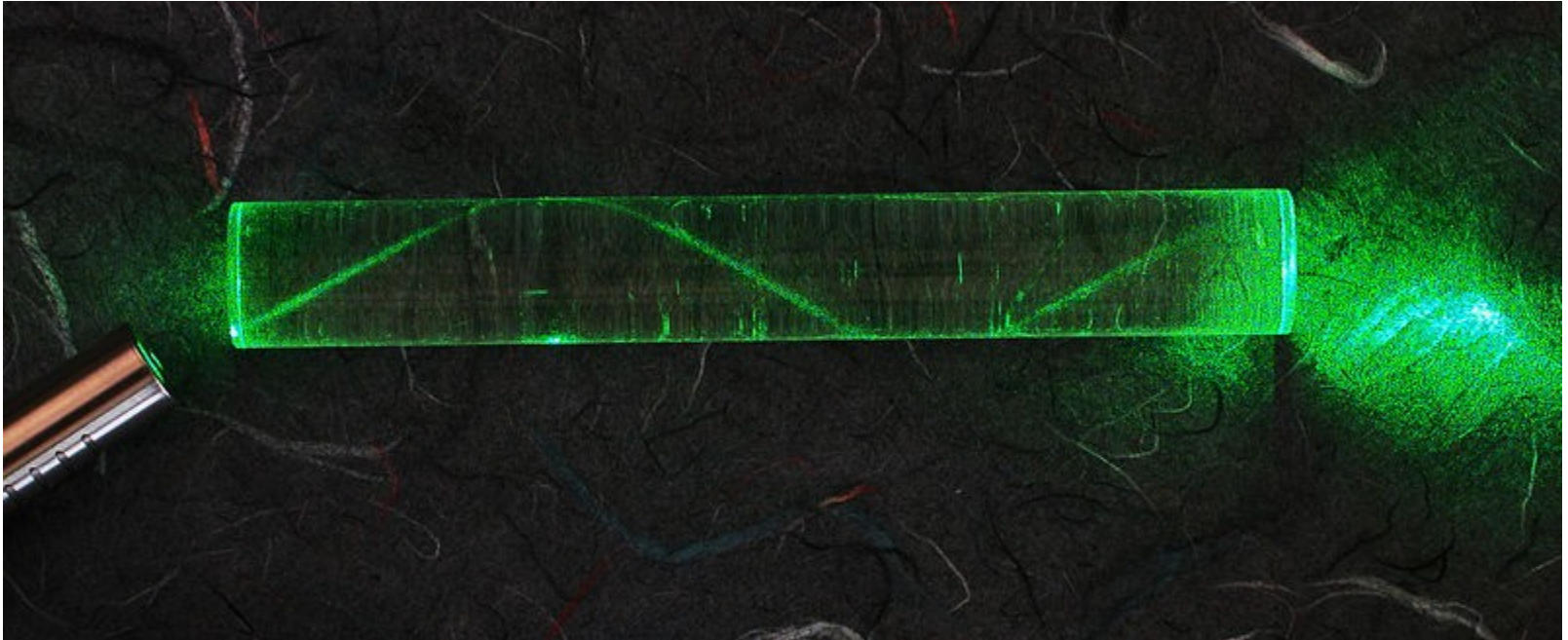
Src:https://commons.wikimedia.org/wiki/File:Optical_fiber_cable.jpg

□ Injection Laser Diode (ILD)

- More efficient, has greater data rates

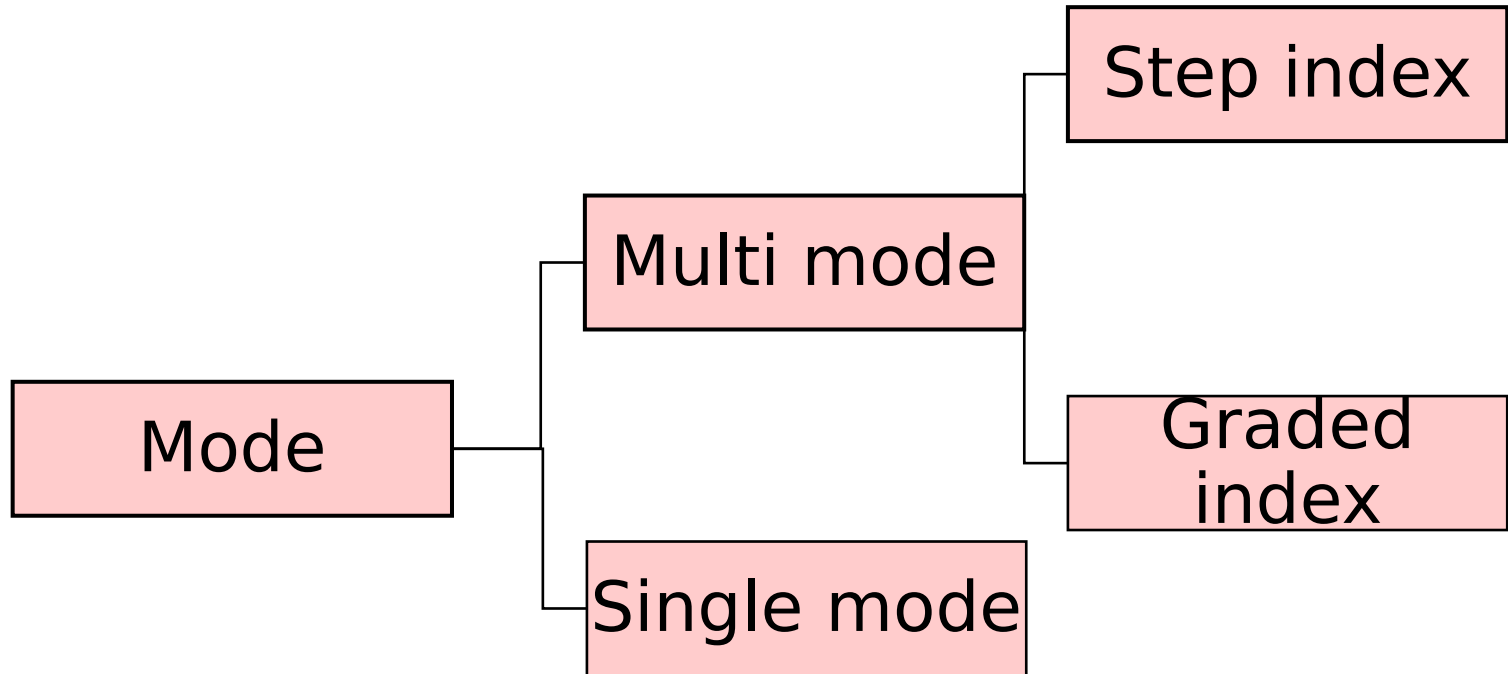


Optical Fiber



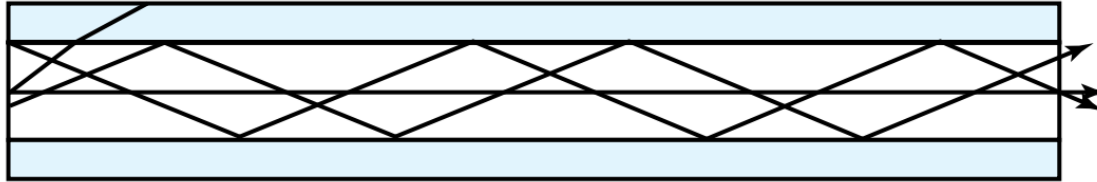
Src:
https://commons.wikimedia.org/wiki/File:Laser_in_fibre.jpg

Optical Fiber

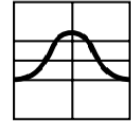


Optical Fiber Transmission Modes

Input pulse

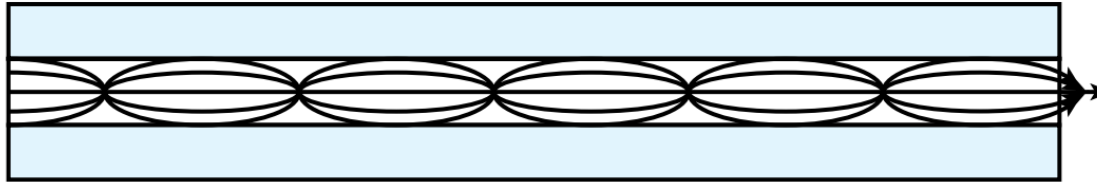
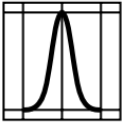


Output pulse

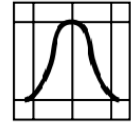


(a) Step-index multimode

Input pulse



Output pulse

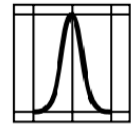


(b) Graded-index multimode

Input pulse

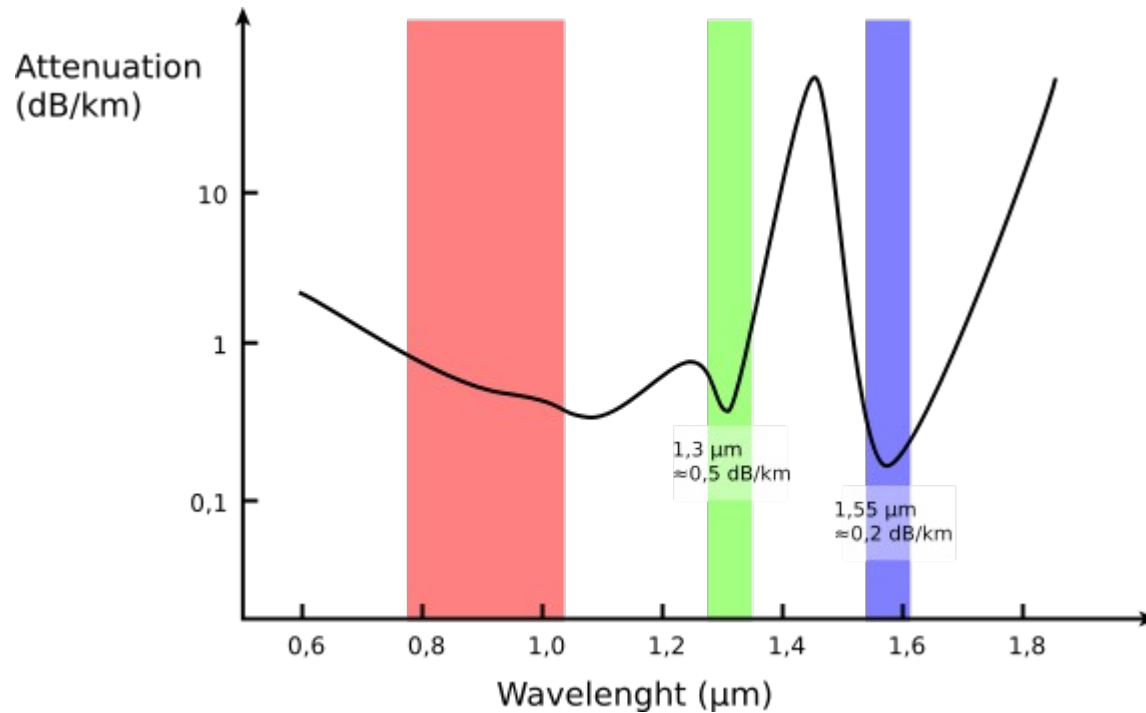


Output pulse



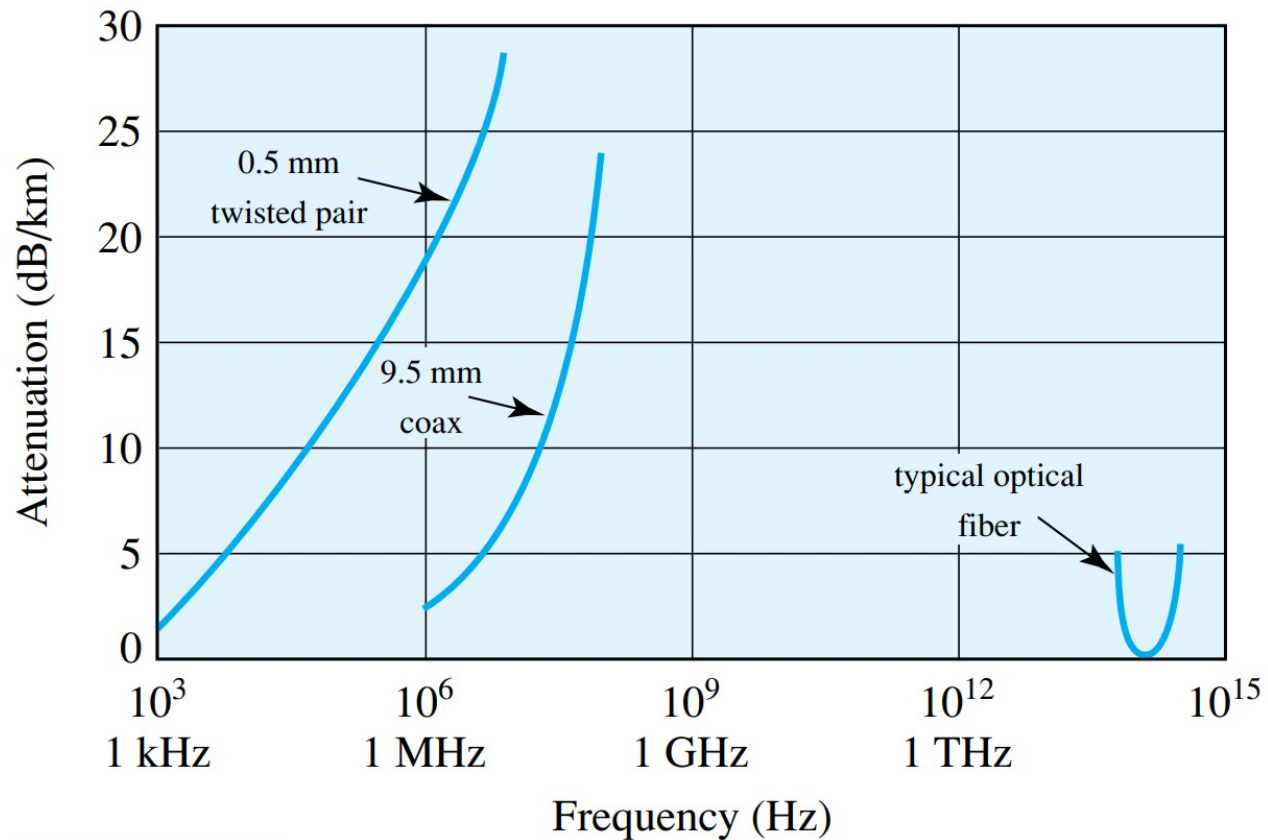
(c) Single mode

Optical Fiber Performance



Src:
https://commons.wikimedia.org/wiki/File:Optical_fiber_transmission_window.svg

Attenuation Comparison



Optical Fiber - Benefits

❑ Greater capacity

- Data rates of 100 Gbps+ (as compared to 1 Gps with electrical cables)

❑ Smaller size and lighter weight

- Considerably thinner than coaxial or twisted pair cable

❑ Lower attenuation

- Maximum distance is 40 km → as compared to 2 km (twisted pair) and 10 km (coaxial cable)

❑ Greater repeater spacing

- Lower cost and fewer sources of error

❑ Electromagnetic isolation

- Not vulnerable to interference, impulse noise, or crosstalk
 - High degree of security from eavesdropping
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Summary

□ Wired transmission medium:

- Twisted pair cable
 - Coaxial cable
 - Optical fiber
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